

Energy Audit Report

Homeowner(s): David Gibson

Address: 105 Eden Street, Bar Harbor, ME 04609

Owner Contact: dgibson@coa.edu

Auditors: David Gibson, Helen Lloyd, Mabon Young

Contact: David Gibson, dgibson@coa.edu, (802) 266-0301

Report Date: June 9, 2025

We conducted an energy assessment of your home on 6/9/2025. This report will tell you what we did, what we found, and what we suggest for your home. These suggestions include information on incentives and financing to make improvements more affordable.



Add here the table of content: in Word go to references and add table of content, in Docs go to the insert and on the bottom add the table of context

Summary of your Audit

Visual Inspection and Measurements

We started with a tour and visual inspection of the inside and outside of the home. We identified any visible damage to the building, moisture control strategies, major appliances, and insulation. We measured square footage and volume of the home, as well as the area of all exterior windows and doors. We used a kill-a-watt meter to measure the electricity use of some appliances. During your audit, we used a carbon monoxide meter to measure the ambient carbon monoxide levels throughout the home.

Attic

We entered the attic to check for insulation, air sealing, ventilation, and potential hazards such as mold. Additionally, we visually inspected the attic ventilation and any duct and pipework passing through the attic.

Basement

We visually inspected any appliances in the basement and noted insulation levels, moisture, rodents, and any other concerns.

Blower Door / Air Leakage Test

A blower door test measures the air tightness of a building. We used a specialized fan in an exterior door to generate negative pressure inside your house. The resulting pressure difference between the inside and outside of the house allows us to measure air leakage. To locate the leaks, we used an infrared camera to check for unusually hot and cold spots. We also checked the pressure differences of the rooms to help determine major air leak locations.

Combustion Appliance Safety

We assessed combustion appliances that burn fossil fuels such as propane, heating oil, or kerosene. These include furnaces, boilers, water heaters, and gas ovens. We visually inspected the combustion appliance(s) in your home, but we were unable to perform combustion safety tests. We also performed gas leak detection tests on your propane appliance(s).

Summary of Recommendations

We recommend the following upgrades for your home. Detailed information about these recommendations and financial resources can be found in other sections of this report. These are organized from the least expensive/most cost-effective and easiest interventions to ones that are more challenging to implement.

What We Found

Basics

Exterior

Interior/Living space

Blower Door / Air Leakage Test

A blower door test simulates a 20mph wind hitting your house from all sides.

To run the test, we used a large fan in an exterior door to depressurize your house. As air is pulled out through the fan, an equal volume of air is pulled in through all of the gaps, cracks, and air leaks throughout the house. This allows us to determine the volume of air leakage into the house and to locate bigger air leaks.

To find leaks, we used an infrared camera to check for unusually hot and cold spots. We also checked the pressure differences of the rooms to help determine major air leak locations.

Air leaks are a big source of heat gain in warm weather and heat loss in cold weather. They also allow moisture to get into the home. Below are some numbers, pictures, and descriptions explaining what we found.



Recommendations	Description
LEDs	Switch your light bulbs to LED light bulbs. LEDs use 80% less energy than incandescent light bulbs which can significantly reduce your electricity bill.
WindowDressers	WindowDressers insulating window inserts help air-seal windows and reduce heat loss and gain.
Gutters	Install gutters and downspouts that divert water at least six feet away from the foundation and to where the ground slopes away from the house.
Attic Air Sealing and Insulation	Air seal the attic and insulate it to at least R-60 (18" of loose-fill cellulose insulation).
Blown-in Cellulose Wall Insulation	Insulate the wall stud cavities with dense packed, blown-in cellulose insulation.
Continuous Exterior Wall Insulation	Add a continuous layer of insulation and potentially replace the air and moisture barrier once it becomes time to replace the siding.
Electrical Panel Upgrade	Consult with an electrician about replacing your existing electrical panel with a 200 amp panel as you add electrical appliances to your home.
Whole-House Surge Protection	Upgrade your electrical panel to add whole-house surge protection.
DIY Air Sealing	Seal easy-to-access air leaks and weatherstrip exterior doors.
Replace Windows	Replace single-paned windows with double- or triple-paned windows and ensure the frames are airsealed. We recommend installing windows that are Energy Star certified for 'Northern' climate.
Replace Doors	Replace uninsulated exterior doors and ensure new doors are insulated and properly weatherstripped. We recommend installing doors that are Energy Star certified for 'Northern' climate.
Other 1	TYPE HERE 1
Other 2	TYPE HERE 2

Info	Values
Date Built:	NA
Foundation Type:	Crawlspace
Attic:	Loose Cellulose , 2 inches
Number of floors	1
Square footage of conditioned space:	202
Volume of conditioned space:	1870
Ambient Carbon Monoxide reading:	0

Info	Values
Roof age:	NA
Orientation:	East/West
Roof type:	Asphalt Shinglesin faircondition. NA
Moisture control:	Current moisture control strategies: NA. These were in NA condition .NA
Siding:	wood siding in fair condition. paint chipping off
Exterior doors:	There are 1 insulated wood exterior doors. In poor condition, totaling 18.3 square feet.
Exterior windows:	There are 4 single paned wood windows. In poor condition, totaling 58 square feet.

Info	Values
Walls:	The framing type of the home is balloon framing, which means the wall cavities extend the full height of the building (from the foundation to the roof). Often the wall cavities are inter-connected with the floor and ceiling cavities. This framing type can create connected passage-ways for air to leave the house and rodents to enter the house. There is Loose Cellulose, Uninsulated insulation 4 inches thick in excellent condition. cellulose not present everywhere
Living room:	NA
Bathroom(s):	NA

Info	Values	Description
CFM50:	910	CFM50 describes how many cubic feet per minute of air are leaving the house at 50 pascals of pressure difference (while the blower door is running). For every cubic foot of air that leaves the house, a cubic foot of air enters the house as well. The higher the number, the leakier the house.
ACH50:	29.2	ACH50 tells us how many air changes per hour are taking place in the house at 50 pascals of pressure difference. This value is normalized for the volume of the house and thus allows for comparison between different houses. The higher the number, the leakier the house.
Equivalent leakage area:	91 under natural conditions. TEST	This is the area (in square inches) equivalent to all of the air leaks in the house combined.
ACHnatural:	1.6	Accounting for the volume of the home, this means that the house exchanges –% of its air every hour. Over one day, the house goes through – complete air changes.

Using a thermal imaging camera, we looked for major air leakage locations and thermal bridging, where heat is bypassing the insulation. There was evidence of

Attic

Basement

Electrical and Mechanical Systems

Energy Bills

Recommendations

These recommendations are organized from the least expensive/most cost-effective and easiest interventions to ones that are more challenging to implement.

LEDs

Problem

Your lighting uses more energy than LED bulbs would. Halogen and incandescent light bulbs use 5-8x more energy than LEDs. This requires significantly more electricity and costs a lot.

Recommendation

Switch your light bulbs to LED light bulbs. LEDs use 80% less energy than incandescent light bulbs which can significantly reduce your electricity bill.

Estimated Cost/Benefits/Incentives

We provide free LED light bulbs; contact us for some if we did not give you any during the audit. Depending on usage, LEDs can save more than \$50 a year.

Window Dressers

Problem

Your windows are a source of heat loss, reduce your comfort due to cold surface temperatures, and may also cause moisture condensation or frost in the winter.

Recommendation

WindowDressers insulating window inserts help air-seal windows and reduce heat loss and gain.

Estimated Cost/Benefits/Incentives

Info Values

Area 202

(sq

ft):

Insulation

type: Cellulose

Insulation

con-

Info	Values
Area (sq ft):	202
Insulation type:	Uninsulated
Insulation condition:	NANA
Insulation of appliances:	No appliances
Moisture Control:	NAin NAcondition. NA
Ducts:	NoNA
Other observations:	There is NANA

Info	Values
Electrical panel:	The electrical panel has an amperage of 30. There are NA unused breaker spaces. no breaker panel, old fuse box in attic

Reconsider this recommendation

Gutters

Problem

The house has no gutters, so water runs off your roof and directly onto the ground next to your house. Water that splashes off the ground can damage the exterior of your house. Runoff that is absorbed into the ground can seep into the basement/crawlspace, causing moisture issues there and throughout your house.

Recommendation

Install gutters and downspouts that divert water at least six feet away from the foundation and to where the ground slopes away from the house.

Estimated Cost/Benefits/Incentives

Adding gutters would mitigate moisture damage and save you from having to replace/fix other parts of the house.

Type	kWh/gallons/cords/tonns	Type2
Propane	90	cost_in_dollars_propane

Attic Insulation and Air Sealing

Problem

The attic is uninsulated or is missing insulation and the attic could also benefit from air-sealing. These upgrades will reduce draftiness and heat loss in the winter and heat gain in the summer.

Recommendation

Air seal the attic and insulate it to at least R-60 (18" of loose-fill cellulose insulation).

Estimated Cost/Benefits/Incentives

Insulation projects are not eligible for federal or state funding in second homes that are not heated year round.

Blow-in cellulose wall insulation

Problem

The walls of the house are uninsulated, which is a significant source of heat loss.

Recommendation

Insulate the wall stud cavities with dense packed, blown-in cellulose insulation. *Estimated Cost/Benefits/Incentives*

Insulation projects are not eligible for federal or state funding in second homes that are not heated year round.

Continuous exterior wall insulation

Problem

There isn't continuous insulation between the siding and the exterior walls of the house. Continuous insulation helps create a complete thermal boundary for your house, and it is now an energy code requirement for new houses.

Recommendation

Add a continuous layer of insulation and potentially replace the air and moisture barrier once it becomes time to replace the siding.

Estimated Cost/Benefits/Incentives

Insulation projects are not eligible for federal or state funding in second homes that are not heated year round.

Electrical Panel Upgrade

Problem

Auditor to describe

Recommendation

Consult with an electrician about replacing your existing electrical panel with a 200 amp panel as you add electrical appliances to your home.

Estimated Cost/Benefits/Incentives

Costs of electrical components needed to support residential energy upgrades – including panels, sub-panels, branch circuits, and feeders – qualify for a 30% nonrefundable tax credit (up to \$600 per item) if they have a capacity of 200 amps or more. An electrical panel upgrades also provides a good opportunity for an electrician to install whole-house surge protection, which costs \$100-\$400. Here are some options: <https://www.popularmechanics.com/home/interior-projects/g43140886/best-whole-house-surge-protectors/>

Whole House Surge Protection

Problem

Your house does not have whole-house surge protection. This will protect your large electrical appliances from damage in the case of an electrical surge. Heat pumps, heat pump water heaters, and other electronics are particularly sensitive to electrical surges.

Recommendation

Seal easy-to-access air leaks and weatherstrip exterior doors. *Estimated Cost/Benefits/Incentives*

The device costs \$100-\$400 and needs to be installed by an electrician. Here are some options: <https://www.popularmechanics.com/home/interior-projects/g43140886/best-whole-house-surge-protectors/>

DIY Air Sealing

Problem

During the blower door test, we noted sources of air leakage in the living areas of your house.

Auditor to detail customized findings.

Recommendation

Air sealing the air leaks we identified around the living areas of your house can be a DIY solution to lower energy use, or an insulation contractor can seal the leaks when they come

to your house for a larger project. Please address any moisture concerns before undertaking significant air sealing efforts.

Estimated Cost/Benefits/Incentives

Ductwork insulation projects are not eligible for federal or state funding in second homes.

Other 1

Problem

Recommendation

Upgrade your electrical panel to add whole-house surge protection.

Estimated Cost/Benefits/Incentives

Auditor to describe. Secondary residence, not heated year-round

Other 2

Problem

Recommendation

TYPE HERE 2

Estimated Cost/Benefits/Incentives

Auditor to describe. Secondary residence, not heated year-round

Replace Windows

Problem

Need to get the description of the problem

Recommendation

Replace single-paned windows with double- or triple-paned windows and ensure the frames are airsealed. We recommend installing windows that are Energy Star certified for ‘Northern’ climate.*Estimated Costs/Benefits/Incentives*

New windows are not eligible for federal or state funding in second homes.

Replace Doors

Problem

Need to get the description of the problem

Recommendation

Replace uninsulated exterior doors and ensure new doors are insulated and properly weatherstripped. We recommend installing doors that are Energy Star certified for ‘Northern’ climate. *Estimated Costs/Benefits/Incentives*

New doors are not eligible for federal or state funding in second homes.

Additional Resources

Many home efficiency upgrades are eligible for Efficiency Maine rebates and federal tax credits. Financial incentives vary based on income level and type of occupancy. An overview of all state rebates and federal tax credits can be found here: <https://www.efficiencymaine.com/at-home/>.

You can search for Efficiency Maine registered contractors here: <https://www.efficiencymaine.com/at-home/vendor-locator/>.

We recommend discussing a whole-home plan for insulation and air sealing work in the basement, walls, and attic with contractors to determine how to best sequence the work to take full advantage of the Efficiency Maine rebates (which are subject to lifetime maximums) and nonrefundable federal tax credits (which are subject to annual maximums).

Efficiency Maine also offers home energy loans to help homeowners who are Maine residents pay for energy and related health and safety upgrades. There are no fees, and interest rates for multi-year loans are between 5.99% (max \$7,500) and 7.99% (max \$25,000), depending on income level. Low- and moderate-income homeowners who borrow the full \$7,500 and pay it back over 10 years will pay a monthly cost of \$83. There is also a 1-year, 0% APR option. To learn more or apply, call 866-376-2463 or visit <https://www.efficiencymaine.com/home-energy-loans/>

Section Four Tests

```
# for (i in 1:length(ranked_sect_4_problem_text)) {  
#   cat(ranked_sect_4_problem_text$Problem)  
# }
```

```

# Create a vector filled with random normal values

name.vec <- as.vector(ranked_sect_4_problem_text$Problem)

name.vec2 <- as.vector(ranked_sect_4_problem_text$Problem)
#u1 <- rnorm(30)
#print("This loop calculates the square of the first 10 elements of vector u1")

# # Initialize `usq`
# usq <- ""

for (i in 1:nrow(ranked_sect_4_problem_text)) {
  # i-th element of `u1` squared into `i`-th position of `usq`

  cat("###", ranked_sect_4_problem_text$Recommendations[i] , "\n")
}

```

LEDs

WindowDressers

Gutters

Attic Air Sealing and Insulation

Blown-in Cellulose Wall Insulation

Continuous Exterior Wall Insulation

Electrical Panel Upgrade

Whole-House Surge Protection

DIY Air Sealing

Replace Windows

Replace Doors

Other 1

Other 2

```
# if( window_dressers == TRUE){  
#   cat(ranked_sect_4_problem_text)  
# }else{  
#   cat("")  
# }  
#  
# if(window_dressers == TRUE){  
#   cat(auto_window_dressers$Description  
#  
#   )  
# }else{  
#   cat("")  
#   }  
#  
# if(window_dressers == TRUE){
```



```

#   cat("
# *Estimated Cost/Benefits/Incentives*
#
# ",auto_window_dresserds_estimates$Description)
# }else{
#   cat("")
# }

df <- data.frame(
  Name = c("Alice", "Bob", "Carol", "Dave", "Eve", "Frank"),
  Score = c(90, 85, 88, 92, 95, 87)
)

# Specific rows to print
rows_to_print <- c(2, 4, 6)

for (i in rows_to_print) {
  cat("Row", i, ": Name =", df$Name[i], ", Score =", df$Score[i], "\n
    ")
}

```

Row 2 : Name = Bob , Score = 85

Row 4 : Name = Dave , Score = 92

Row 6 : Name = Frank , Score = 87

```

# cat("###", ranked_sect_4_problem_text$Recommendations[i] , "\n"
#   ,
#
#   "*Problem*", i ,"\n")

```